# Chapter 3

# Modeling and Parameter Structure

A quadrotor that picks up a parcel does not merely become heavier. It also becomes harder to rotate: on the platform of this work, a 0.30 kg parcel hanging about a quarter of a metre below the rotor plane adds fifteen percent to the mass but more than doubles the roll and pitch inertia (Section 3.3.2). These are two distinct changes to the equations of motion, and a controller that compensates only the first loses inner-loop bandwidth in a way that mass adaptation cannot explain.

This chapter derives the model that the estimator and the controller share and makes explicit which of its parameters are identified online. Its structural claim is that, for a payload grasped on the thrust axis, both changes are governed by a *single* online scalar. The composite mass $m_T$ is identified from the translational channel; the roll and pitch inertia increment then follows from $m_T$ by rigid-body algebra and the nominal vertical arm of the gripper. No payload mass is supplied to either optimiser, and no attach or release notification to the estimator. The nominal arm is a geometric prior, and Section 3.4.4 states exactly what that prior is and what it is not.

Section 3.1 defines the system and its assumptions, Section 3.2 its dynamics, and Section 3.3 the coupling between payload mass and inertia. Section 3.4 reduces the payload to one online degree of freedom and explains how the estimator treats the rotational channel. Section 3.5 describes the measured input

and the interfaces between estimator, controller and autopilot, and Section 3.6 collects the modelling assumptions and their limits. The numerical form of the two optimisation problems, their initialisation and constraints, the residual specific force and the inner-loop scheduling belong to Chapter 4; the experimental evidence for the design choices belongs to Chapter 5.

## 3.1 System Definition and Modelling Assumptions

Two reference frames are used throughout. The world frame $\mathscr{W}$ is East–North–Up (ENU), so that gravity acts along $-z$. The body frame $\mathscr{B}$ is Front–Left–Up (FLU) with its origin at the geometric centre of the rotor disc — *not* at the centre of mass. The thrust vector passes through the body origin by construction.

The state vector and the control input are

$$x = \left[\boldsymbol{p}; \boldsymbol{v}; \boldsymbol{q}; \boldsymbol{\omega}\right] \in \mathbb{R}^{13}, \qquad u = \left[T; \boldsymbol{\tau}\right] \in \mathbb{R}^{4}, \tag{3.1}$$

where $\boldsymbol{p}$ and $\boldsymbol{v}$ are the position and velocity in $\mathscr{W}$, $\boldsymbol{q}$ is the unit quaternion encoding the attitude of $\mathscr{B}$ relative to $\mathscr{W}$ (Section 3.1.1), $\boldsymbol{\omega}$ is the body rate in $\mathscr{B}$, $T$ is the collective thrust along the body $z$ axis, and $\boldsymbol{\tau}$ is the body torque. We abbreviate $\boldsymbol{e}_3 = [0,0,1]^\top$, and write $m_B$, $m_P$ and $m_T = m_B + m_P$ for the bare-airframe, payload and composite masses.

Three assumptions define the system.

1. *The bare airframe is balanced about the body origin*, so that its centre of mass and the body origin coincide when no payload is attached. This is an assumption about the vehicle, not a consequence of the frame choice.

2. *The payload is a point mass $m_P$ rigidly attached on the thrust axis*, at the body-

frame offset $\boldsymbol{r}_p = [0, 0, r_z]^\top$ with $r_z < 0$: the load hangs below the airframe. The gripper aligns the vehicle over the parcel before attaching; the residual horizontal offset that alignment leaves is treated as an approximation in Section 3.6.

3. *Rotor thrust follows the static quadratic law* of Section 3.5, with calibrated coefficients.

### 3.1.1 Attitude representation

Attitude is represented by a unit quaternion under the *Hamilton* convention with the scalar part first,

$$\boldsymbol{q} = [q_w, q_x, q_y, q_z]^\top, \qquad \|\boldsymbol{q}\| = 1, \tag{3.2}$$

encoding the rotation from $\mathscr{B}$ to $\mathscr{W}$. The corresponding rotation matrix is

$$R(\boldsymbol{q}) = \begin{bmatrix} q_w^2 + q_x^2 - q_y^2 - q_z^2 & 2(q_x q_y - q_w q_z) & 2(q_x q_z + q_w q_y) \\ 2(q_x q_y + q_w q_z) & q_w^2 - q_x^2 + q_y^2 - q_z^2 & 2(q_y q_z - q_w q_x) \\ 2(q_x q_z - q_w q_y) & 2(q_y q_z + q_w q_x) & q_w^2 - q_x^2 - q_y^2 + q_z^2 \end{bmatrix}. \tag{3.3}$$

Three conventions are fixed here and used throughout: the Hamilton product, the scalar-first component order of (3.2), and $\boldsymbol{\omega}$ resolved in $\mathscr{B}$.

One scalar derived from (3.3) is used repeatedly: the fraction of the collective thrust that acts vertically. With roll $\phi$ and pitch $\theta$ in the ZYX Euler convention $R = R_z(\psi)R_y(\theta)R_x(\phi)$, it is the $(3,3)$ entry of the rotation matrix,

$$\cos\phi\,\cos\theta \;=\; \boldsymbol{e}_3^\top R(\boldsymbol{q})\,\boldsymbol{e}_3 \;=\; 1 - 2\big(q_x^2 + q_y^2\big). \tag{3.4}$$

The implementation evaluates the right-hand side directly from the quaternion.

## 3.2 Quadrotor Dynamics with a Rigid Payload

### 3.2.1 Kinematics

Translational kinematics is the definition of velocity,

$$\dot{\boldsymbol{p}} = \boldsymbol{v}. \tag{3.5}$$

With the body rate resolved in $\mathscr{B}$, attitude kinematics is the Hamilton product of the attitude with the pure quaternion of the body rate,

$$\dot{\boldsymbol{q}} = \tfrac{1}{2}\,\boldsymbol{q} \otimes [0, \boldsymbol{\omega}] = \tfrac{1}{2}\,\Xi(\boldsymbol{q})\,\boldsymbol{\omega}, \qquad \Xi(\boldsymbol{q}) = \begin{bmatrix} -q_x & -q_y & -q_z \\ q_w & -q_z & q_y \\ q_z & q_w & -q_x \\ -q_y & q_x & q_w \end{bmatrix}. \tag{3.6}$$

The body rate enters on the *right* of the product because it is resolved in the body frame; a world-frame rate would enter on the left. The implementation uses the matrix form, which is linear in $\boldsymbol{\omega}$. One property of $\Xi$ matters for estimation: $\boldsymbol{q}^\top \Xi(\boldsymbol{q}) = \mathbf{0}^\top$ for every quaternion, so $\boldsymbol{q}^\top \dot{\boldsymbol{q}} = 0$ and the kinematics preserve $\|\boldsymbol{q}\|$ exactly. The estimator integrates (3.6) across a whole window before comparing it with measurements, and any drift of the norm would appear as an attitude residual that the optimiser could attribute to the payload parameter.

Neither (3.5) nor (3.6) contains a mass, an inertia, or any other physical parameter. *The payload therefore enters the model only through the dynamics.*

### 3.2.2 Translational dynamics

Newton's second law in $\mathscr{W}$, with thrust rotated from the body frame and a linear aerodynamic drag term, gives

$$\dot{\boldsymbol{v}} = \frac{1}{m_T}\Big(R(\boldsymbol{q})\,\boldsymbol{e}_3\,T - k_d\,\boldsymbol{v}\Big) - g\,\boldsymbol{e}_3, \tag{3.7}$$

where $m_T$ is the *composite* mass of airframe plus payload, $k_d$ is a fixed drag coefficient, and $g$ is the gravitational acceleration. The mass — and only the mass — enters here, as the scalar gain $1/m_T$ from thrust to acceleration. If the thrust is known independently of the model, the measured motion determines $m_T$ through (3.7) in ordinary near-hover flight, with no dedicated excitation manoeuvre; the qualification "independently of the model" is the subject of Section 3.5.

Projecting (3.7) onto $\boldsymbol{e}_3$ and neglecting the vertical component of the drag term gives the vertical force balance

$$m_T\big(g + a_z\big) \;=\; T\cos\phi\,\cos\theta, \tag{3.8}$$

with $\cos\phi\cos\theta$ from (3.4) and $a_z = \dot{v}_z$. It is used to initialise the estimator (Section 4.2.3).

### 3.2.3 Rotational dynamics

Let $J_B = \mathrm{diag}(J_{xx}, J_{yy}, J_{zz})$ denote the inertia of the bare airframe about its own centre of mass, and $J_T$ the composite inertia derived in Section 3.3.1. Euler's equation in the body frame is

$$\dot{\boldsymbol{\omega}} = J_T^{-1}\Big(\boldsymbol{\tau} - \boldsymbol{\omega}\times J_T\boldsymbol{\omega}\Big), \tag{3.9}$$

where $\boldsymbol{\omega} \times J_T \boldsymbol{\omega}$ is the gyroscopic coupling. Equation (3.9) holds in this form only when the inertia and the applied moments refer to the same point, and for a free-flying body that point must be the centre of mass. The composite inertia must therefore be taken about the composite centre of mass, which *moves* when a payload is attached (Section 3.3.1). For the centred payload of assumption 2 that centre of mass stays on the thrust axis, so thrust contributes no moment to (3.9).

Taken together, (3.7) and (3.9) give each payload-dependent quantity exactly one point of entry: the composite mass appears only as $1/m_T$ in the translational dynamics, and the inertia increment only inside $J_T$ in the rotational dynamics. This separation is what allows the two effects to be attributed to distinct physical channels, and it is used throughout Section 3.4.

## 3.3 Payload-Induced Mass and Inertia Coupling

With the payload of assumption 2, define the composite mass and the *reduced mass*

$$m_T = m_B + m_P, \qquad \mu = \frac{m_B m_P}{m_B + m_P} = \frac{m_B m_P}{m_T}. \tag{3.10}$$

### 3.3.1 Inertia increment and the reduced mass

The inertia increment requires more care than the parallel-axis theorem applied naively, because *the centre of mass itself moves when the payload is attached*. With the airframe centre of mass at the body origin, the composite centre of mass lies on the thrust axis at

$$\boldsymbol{c} = \frac{m_P}{m_T} \boldsymbol{r}_p = \left[0,\, 0,\, \frac{m_P}{m_T} r_z\right]^\top. \tag{3.11}$$

The theorem must be applied about this *new* centre of mass, and both bodies contribute, since the airframe no longer sits at the centre of mass either.

Consider a roll or pitch axis and let $d = |r_z|$ denote the lever arm. From (3.11), the distances of the two bodies from the composite centre of mass are

$$\underbrace{\|\mathbf{0} - \boldsymbol{c}\| = \frac{m_P}{m_T} d,}_{\text{airframe}} \qquad \underbrace{\|\boldsymbol{r}_p - \boldsymbol{c}\| = \Big(1 - \frac{m_P}{m_T}\Big) d = \frac{m_B}{m_T} d\,.}_{\text{payload}} \tag{3.12}$$

The payload's distance is *smaller* than $d$. Adding the two parallel-axis contributions and simplifying,

$$\begin{aligned} \Delta J &= m_B \Big(\frac{m_P}{m_T} d\Big)^2 + m_P \Big(\frac{m_B}{m_T} d\Big)^2 \\ &= \frac{d^2}{m_T^2}\Big(m_B m_P^2 + m_P m_B^2\Big) = \frac{d^2}{m_T^2} m_B m_P \big(m_P + m_B\big) \\ &= \frac{m_B m_P}{m_T} r_z^2 = \mu\, r_z^2. \end{aligned} \tag{3.13}$$

The factor $(m_B + m_P)$ produced by the sum cancels one power of $m_T$ in the denominator, and the two contributions collapse onto the reduced mass (3.10) — the same quantity that governs the two-body problem, and for the same reason: the inertia of a rigid pair about its centre of mass depends only on the *relative* position of the two bodies.

Equation (3.13) also identifies a natural error. Writing $\Delta J = m_P r_z^2$ — treating the payload as orbiting a fixed body origin, and forgetting both the airframe's own contribution and the displacement of the centre of mass — overestimates the increment by the factor $m_T/m_B$, which is 14.5 % at the heavier payload of this work.

The same computation carried out for the full tensor gives, for an on-axis point mass,

$$J_T \;=\; J_B + \mu\Big(\|\boldsymbol{r}_p\|^2 I_3 - \boldsymbol{r}_p \boldsymbol{r}_p^{\top}\Big) \;=\; J_B + \operatorname{diag}\big(\mu r_z^2,\, \mu r_z^2,\, 0\big). \tag{3.14}$$

Three consequences are used below. The increment is *isotropic* in roll and pitch, so a single scalar $\Delta J = \mu r_z^2$ describes both. It is *diagonal*: every off-diagonal product of inertia contains a horizontal arm and vanishes. And the yaw increment is exactly zero for a point mass on the yaw axis; the parcel's own rotational inertia about its centre is the only yaw contribution, and Section 3.6 declares its omission.

### 3.3.2 Magnitudes on the experimental platform

The factor by which the payload slows the roll and pitch response to a given torque is the inertia ratio

$$\kappa = \frac{J_{xx} + \Delta J}{J_{xx}}. \tag{3.15}$$

For the two payloads used in the experiments, attached at the nominal arm $r_z = -0.27\,\mathrm{m}$ on the platform of this work ($m_B = 2.0643\,\mathrm{kg}$, $J_{xx} = 0.0142\,\mathrm{kg\,m^2}$), equations (3.10), (3.13) and (3.15) give

| | $\mu$ [kg] | $\Delta J$ [kg m$^2$] | $\kappa$ |
|---|---|---|---|
| $m_P = 0.15\,\mathrm{kg}$ | 0.140 | 0.0102 | 1.72 |
| $m_P = 0.30\,\mathrm{kg}$ | 0.262 | 0.0191 | 2.34 |

(3.16)

A 0.30 kg parcel on a 2.06 kg airframe is a 15 % change of mass but a more than *two-fold* change of roll and pitch inertia. Adapting the mass alone, in the thrust channel, accounts for the first and not the second. The increment is governed by how far *below* the airframe the load hangs, through $r_z^2$, which is why the gripper geometry, and not only the parcel, sets the difficulty of the task.

## 3.4 Single-Parameter Estimation Structure

### 3.4.1 One online degree of freedom

Equations (3.10)–(3.14) express every payload-dependent quantity of the model through two numbers, the payload mass $m_P$ and the vertical arm $r_z$, and the two are not on an equal footing.

The mass changes from flight to flight and within a flight: it is zero before a grasp, takes the parcel's value after it, and returns to zero at release, possibly in several steps if a load is lost piecewise. It is also *observable*, through the gain $1/m_T$ of (3.7). It is therefore the quantity estimated online, in the form of the composite mass $m_T$, and the payload mass follows as

$$m_P = (m_T - m_B)^+, \tag{3.17}$$

where $(\cdot)^+$ denotes the positive part, because the mass estimate is permitted to fall slightly below $m_B$ to leave room for calibration error.

The vertical arm, by contrast, is a property of the gripper rather than of the parcel. It is fixed by where the gripper holds the load relative to the rotor plane, and it enters the equations only through the inertia increment, which is weakly excited in delivery flight. It is therefore held at a nominal value rather than estimated. The inertia increment consumed by the model is then a function of the single online scalar,

$$\boxed{\Delta J(m_T) \;=\; \frac{m_B\,(m_T - m_B)^+}{m_T}\, r_z^2} \tag{3.18}$$

which vanishes when the mass estimate returns to the bare-airframe value. No attach or release event is needed to switch the inertia on or off: the algebra does it, continuously, as the estimate moves. The mass is estimated; the inertia

is derived from it and is never an independent decision variable.

### 3.4.2 Which channel identifies the mass

Equation (3.18) makes the rotational dynamics depend on $m_T$. Should the *estimator* also be allowed to use that dependence, so that the angular-rate residual contributes information about the mass?

In principle it should: the increment is larger than the airframe inertia itself (Section 3.3.2), so any roll or pitch manoeuvre carries a strong mass signature. In practice the rotational channel of this platform is not modelled well enough to carry an absolute quantity. Three independent errors were found in it: the calibrated airframe inertia differs substantially from the simulator's model description; rotor drag acting through the vertical offset of the rotor plane produces a velocity-dependent body moment that (3.9) does not contain; and the reconstructed torque of an unloaded vehicle carries a small constant bias (Section 3.6). None prevents the vehicle from flying, but an optimiser that is allowed to move $m_T$ inside the rotational model will use the mass to absorb all three. Development flights with that freedom produced large mass overestimates, reported in Section 5.7.2.

The deployed estimator therefore removes the direct sensitivity of the rotational dynamics to the mass decision variable,

$$\left.\frac{\partial \dot{\omega}}{\partial m_T}\right|_{\text{estimator}} = 0, \tag{3.19}$$

so that the mass enters the estimator's model only through (3.7), where the model — a thrust, a mass and gravity — is trustworthy. It does not, however, leave the payload inertia out of its rotational model: an estimator that integrates the bare-airframe inertia sees a heavy, low-hanging load as an unexplained angular response. The increment is instead inserted as a *known parameter* com-

puted from the previous solution $\hat{m}^-$ and held constant within the window,

$$\Delta J^{\mathrm{lag}} = \frac{m_B \left(\hat{m}^- - m_B\right)^+}{\hat{m}^-} \, r_z^2, \qquad \left.\frac{\partial \Delta J^{\mathrm{lag}}}{\partial m_T}\right|_{\mathrm{window}} = 0, \tag{3.20}$$

refreshed between solves at the 10 Hz window rate. In the implementation the positive part in (3.20) is smoothed over a width of 0.02 kg, which near $\hat{m}^- = m_B$ changes the increment by less than the airframe calibration error. Table 3.1 places this *lagged* treatment between the two alternatives.

**Table 3.1: Treatments of the inertia increment inside the estimator. The controller always uses** (3.18) **evaluated at the published estimate; only the estimator's own rotational model differs between rows.**

| Treatment | $\Delta J$ inside the estimator | $\partial\dot{\omega}/\partial m_T$ | Status |
|---|---|---|---|
| omitted | 0 | zero | available, off |
| lagged (deployed) | (3.18) at the previous solution $\hat{m}^-$ | zero | default |
| live | (3.18) at the decision variable $m_T$ | non-zero | rejected |

The corresponding implementation settings are listed in Appendix A. The lagged treatment is the default of the experiment entry points; the library default of the estimator node remains the omitted treatment.

The omitted treatment satisfies (3.19) with the wrong inertia; the live treatment has the right inertia but gives the optimiser the mass as a rotational knob. The lagged treatment keeps both properties: within a solve the increment is a constant, and from one solve to the next it follows the mass. Section 5.7.2 compares the omitted and lagged treatments in closed loop.

Equation (3.19) removes the *direct* sensitivity only, and it should not be read as saying that the rotational channel has no influence on the mass estimate. Two indirect couplings remain. Within a window, the angular-rate residual shapes the estimated attitude trajectory, and the attitude sets the direction of the thrust in (3.7); an error in the rotational model therefore reaches the mass through the attitude. Between windows, (3.20) feeds the previous mass back into the rotational model, so a biased estimate changes the inertia used in the next window. The lagged treatment does not remove these paths; it removes the optimiser's ability to use the mass directly as a free rotational parameter.

### 3.4.3 Estimator formulation

The moving-horizon estimator augments the composite mass into the state vector,

$$x_{\text{aug}} = \left[x;\, m_T\right] \in \mathbb{R}^{14}, \qquad \dot{m_T} = 0, \tag{3.21}$$

holding it constant across the horizon so that a window of measurements constrains a single value rather than a free trajectory. Process noise is admitted on the thirteen physical states only; noise on the mass channel would let it drift within the window and dissolve the constraint that makes the estimate meaningful.

The estimator's runtime parameter vector carries the reconstructed wrench of Section 3.5 and a three-element slot of known geometry,

$$p_{\text{MHE}} = \left[T_{\text{phys}};\, \boldsymbol{\tau}_{\text{phys}};\, \Delta J^{\text{est}};\, c_x^{\text{est}};\, c_y^{\text{est}}\right] \in \mathbb{R}^7, \tag{3.22}$$

which enters the estimator's rotational model as a fixed inertia increment and a fixed thrust-offset moment. In the deployed configuration the first entry carries (3.20) and the two offsets are zero; the omitted treatment sets all three to zero. In neither case is the slot a function of the decision variable, which is how (3.19) is enforced.

The estimator's stage cost is built on one measurement residual. Writing $\boldsymbol{y}_i$ for the odometry measurement at stage $i$ of the window and $x_i$ for the physical part of the estimator's trajectory at that stage,

$$\boldsymbol{e}_i \;=\; \boldsymbol{y}_i - x_i \;=\; \left[\boldsymbol{e}_i^{p};\;\; \boldsymbol{e}_i^{v};\;\; \boldsymbol{e}_i^{q};\;\; \boldsymbol{e}_i^{\omega}\right] \in \mathbb{R}^{13}. \tag{3.23}$$

The *velocity residual* and the *angular-rate residual* referred to throughout are the blocks $\boldsymbol{e}^{v}$ and $\boldsymbol{e}^{\omega}$ of this one vector. It contains no acceleration: the trajectory $x_i$ is produced by integrating the dynamics forward under the known input, so the comparison is between velocities, never between a measured and a modelled

acceleration. Differentiating the odometry velocity would amplify its noise by $1/\Delta t$, to the order of the very effect being estimated.

By (3.19), the mass is identified through the velocity residual: it is the window-constant value that reconciles the reconstructed thrust with the observed velocity trajectory. The angular-rate residual constrains the attitude and rate trajectories and reaches the mass only through the indirect paths described above. Chapter 4 gives the numerical form of the optimisation, its weights, bounds and initialisation.

### 3.4.4 What is estimated, what is derived, and what is fixed

Table 3.2 records the identity of every parameter of the model. One scalar is an online degree of freedom; everything else is derived from it by the algebra of Section 3.3, calibrated once for the airframe, or held at a nominal value.

**Table 3.2: Model parameters, classified by identity. Only the composite mass is a decision variable of the estimator; the inertia increment is generated from it.**

| Symbol | Meaning | Dim. | Identity | Value / enters |
|---|---|---|---|---|
| $m_T$ | composite mass | 1 | **MHE decision variable** | (3.7) |
| $\Delta J$ | roll/pitch inertia increment | 1 | derived, (3.18); lagged in the estimator | (3.26), (3.20) |
| $m_P$ | payload mass | 1 | derived, $(m_T - m_B)^+$ | (3.18) |
| $r_z$ | payload vertical arm | 1 | nominal gripper geometry | $-0.27\,\mathrm{m}$ |
| $\boldsymbol{c}_{xy}$ | CoM horizontal offset | 2 | **not modelled**; held at $\mathbf{0}$ | — |
| $m_B$ | bare-airframe mass | 1 | calibration | $2.0643\,\mathrm{kg}$ |
| $J_B$ | bare-airframe inertia | 3 | calibration | $\mathrm{diag}(0.0142, 0.0142, 0.0210)\,\mathrm{kg\,m^2}$ |
| $k_f, k_m$ | thrust and yaw-reaction coefficients | 2 | calibration | $8.55 \times 10^{-6}\,\mathrm{N\,s^2}$, $0.016\,\mathrm{m}$ |
| $k_d$, $g$ | drag coefficient, gravity | 2 | constants | $0.05\,\mathrm{kg\,s^{-1}}$, $9.81\,\mathrm{m\,s^{-2}}$ |

Three entries of the table invite objection.

**Why not augment the inertia increment as well?** Because it is nearly unobservable in the flight regime of interest, and because the channel through which it would be observed is the one this platform models least well. The increment acts only through $\dot{\omega}$, which a delivery task spent largely in near-hover, low-rate flight excites weakly; a free $\Delta J$ would let the optimiser explain measurement noise and the rotational model errors of Section 3.4.2 with a quantity the data barely constrain. Moreover $J_T$ appears *inverted* in (3.9) and is propagated across the horizon, so an error in it is amplified rather than damped, and inertia elements are two orders of magnitude smaller than the mass, which would worsen the scaling of the estimator's Hessian.

**Why not estimate a horizontal offset?** Because at the operating condition of this work there is none to estimate, and carrying one is costly. The gripper aligns over the parcel before attaching, so the offset is bounded by the alignment tolerance; an estimated offset would be identifiable only through the rotational residual, the channel that Section 3.4.2 keeps away from absolute quantities; and an offset is undefined once the payload has gone, so any parameterisation that carries one must also say what it becomes at release.

**Why is a nominal $r_z$ acceptable?** Because the arm is a property of the gripper, fixed by its design, and not information about the parcel. It is a geometric prior, and the method does use it: it enters the inertia increment (3.18) of both optimisers, where its error propagates quadratically — a 10 % error in $r_z$ is a 21 % error in $\Delta J$, which mis-sizes the predicted attitude response and the inner-loop ratio $\kappa$. It cannot enter the mass estimate through a derivative, because of (3.19), but it can through the indirect paths of Section 3.4.2; the size of that effect has not been measured. The difference between the nominal and the realised arm is quantified in Section 3.6.

**The information boundary.** The precise claim of this dissertation is that *no task-specific payload mass enters the estimator or the controller, and no attach or release notification enters the estimator*. It is not the stronger claim that the flight stack contains no payload-related constant. Two such constants remain and are disclosed: the nominal arm $r_z$ above, and the platform's 0.30 kg payload *envelope*, which sets the inertia floor of the attach-time bootstrap (Section 4.6.3). Both are specifications of the airframe and its gripper, in the same category as $J_B$ or the actuator envelope, and neither is the mass of the parcel carried on a particular flight. The controller's own release command is used to start the confirmation timeout, to end the inertia floor and to gate the reset of the residual force (Chapter 4); it does not set the payload mass.

## 3.5 Measured-Input Reconstruction and Model Interfaces

### 3.5.1 The physical input

The thrust and torque in (3.7) and (3.9) must be known to the estimator. They are reconstructed from measured rotor speeds $\omega_i$ through the quadratic rotor law $F_i = k_f \omega_i^2$ and the linear map from the four rotor thrusts to the wrench. For the X configuration of this platform, with rotor positions $(x_i, y_i)$, $|x_i| = |y_i| = l = 0.174\,\text{m}$, and spin directions $d_i \in \{\pm 1\}$,

$$\begin{bmatrix} T_{\text{phys}} \\ \tau_{\text{phys}} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 & 1 & 1 \\ -l & l & l & -l \\ -l & l & -l & l \\ -k_m & -k_m & k_m & k_m \end{bmatrix}}_{M} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix}, \qquad F_i = k_f \omega_i^2, \tag{3.24}$$

whose rows are $T_{\text{phys}} = \sum_i F_i$, $\tau_x = \sum_i y_i F_i$, $\tau_y = -\sum_i x_i F_i$ and $\tau_z = -k_m \sum_i d_i F_i$, with $k_f = 8.55 \times 10^{-6}\,\text{N s}^2$ and yaw-reaction coefficient $k_m = 0.016\,\text{m}$. The first two moment components are lever-arm products; the third is the aerodynamic reaction torque, proportional to rotor thrust and opposed to the rotor's own spin.

**This is a wrench, not a state estimate.** Equation (3.24) returns the wrench the *actuators generate*. It excludes $J_T \dot{\omega}$ and rotor gyroscopic terms, which is what makes it usable as the known input $u^{\text{phys}}$ of the estimator rather than as a second, competing estimate of the state. Rotor speeds are sampled at 250 Hz and averaged over each 10 Hz estimation interval, consistent with a zero-order-hold discretisation of (3.7) and (3.9).

**Why the commanded input cannot be used.** Let the controller compute its thrust command from an assumed mass $\tilde{m}$, so that in hover $T_{\text{cmd}} = \tilde{m} g$. Feeding $T_{\text{cmd}}$ into (3.7) and solving for the mass yields $m_T = T_{\text{cmd}}/g = \tilde{m}$ for *any* $\tilde{m}$: the relation is satisfied identically, the residual carries no information, and the estimate would reproduce whatever the controller already believed. Any inversion error, saturation or motor lag between command and rotor is likewise invisible. Equation (3.24) breaks this circularity because rotor speeds are a physical measurement into which no model state enters, taken downstream of allocation and the motor dynamics. The principle recurs throughout the design: an adaptive loop must be fed by a signal that is not itself a product of the adaptation. The difference is not academic; the executed yaw torque recorded in flight is about a third of the intended value (Section 5.1).

### 3.5.2 The controller's model

The predictive controller carries the payload parameters as runtime values,

$$p_{\mathrm{NMPC}} = \left[ x_{\mathrm{ref}}(13);\, m_T;\, \Delta J;\, \boldsymbol{c}_{xy}(2);\, \hat{\boldsymbol{d}}(3);\, \boldsymbol{\xi}(3) \right] \in \mathbb{R}^{23}, \tag{3.25}$$

where $m_T$ is the published mass estimate, $\Delta J$ is (3.18) evaluated at it, the two horizontal-offset entries $\boldsymbol{c}_{xy}$ are held at zero (assumption 2), $\hat{\boldsymbol{d}}$ is the residual specific force of Section 4.5.3, and the rotational disturbance $\boldsymbol{\xi}$ is held at zero. The zero entries remain only so that the generated solver is shared with the comparison controllers. The prediction model is the model of Section 3.2 with these parameters substituted,

$$\begin{aligned}
&\dot{\boldsymbol{p}} = \boldsymbol{v}, \qquad \dot{\boldsymbol{q}} = \tfrac{1}{2}\,\Xi(\boldsymbol{q})\,\boldsymbol{\omega}, \\
&\dot{\boldsymbol{v}} = \frac{1}{m_T}\Big( R(\boldsymbol{q})\,\boldsymbol{e}_3\,T - k_d\,\boldsymbol{v} \Big) - g\,\boldsymbol{e}_3 + \hat{\boldsymbol{d}}, \\
&\dot{\boldsymbol{\omega}} = J_T^{-1}\Big( \boldsymbol{\tau} - \boldsymbol{\omega} \times J_T\,\boldsymbol{\omega} \Big), \quad J_T = \mathrm{diag}\big(J_{xx} + \Delta J,\, J_{yy} + \Delta J,\, J_{zz}\big).
\end{aligned} \tag{3.26}$$

Apart from the additive residual $\hat{\boldsymbol{d}}$, nothing in (3.26) is new: the pair $(m_T, \Delta J)$ are the only payload quantities that change from one solve to the next, each enters at exactly one place, and the second is a function of the first.

The published mass is not the raw estimate. The estimator publishes $m_B + m_P^{\mathrm{out}}$, where $m_P^{\mathrm{out}}$ is the positive part (3.17) passed through a first-order filter, so the published mass never falls below $m_B$; the controller filters and rate-limits it again (Section 4.3.2). The controller's and the estimator's inertia increments are therefore evaluated at different points: the controller at the filtered, published estimate, the estimator at its own previous solution. In neither optimiser is the inertia a function of a decision variable, and in both it follows the mass from one solve to the next. The estimator has no counterpart of $\hat{\boldsymbol{d}}$, so a persistent residual cannot be used there to explain away a mass change.

### 3.5.3 The interface to the autopilot

The deployed system is a cascade: the predictive controller is the outer loop, and the autopilot's body-rate controller and mixer are the inner loop. Three torque-like quantities exist in it and must not be confused.

1. The *predicted torque* $\tau$ is a decision variable of the controller's problem. It shapes the predicted state trajectory through (3.26) and never leaves the solver.

2. The *body-rate setpoint* is what the autopilot receives. After each solve the controller publishes, at 50 Hz, a collective thrust and a body-rate setpoint interpolated along the latest predicted trajectory: the thrust from the predicted input sequence, converted to a normalised throttle by (4.26), and the body rate from the predicted state sequence one stage ahead, starting at its second node. The roll and pitch rate setpoints are then scaled for the payload inertia (Section 4.6). The autopilot's rate loop and mixer turn this setpoint into rotor commands.

3. The *physical torque* $\tau_{\text{phys}}$ is what the rotors actually produced, reconstructed from measured rotor speeds by (3.24). It is the only torque the estimator sees.

The division is deliberate: the rate loop runs at a bandwidth the 20 Hz solver cannot match, and leaving allocation on the autopilot keeps saturation handling next to the hardware. It is also why the estimator evaluates $M$ in the *forward* direction from measured rotor speeds rather than reading $\tau$ off the solver: $M$ applied to what the motors did is a measurement, whereas $\tau$ from the solver is an intention. The ROS layer never evaluates the inverse allocation; its role in sizing the controller's input bounds is discussed with the actuator envelope in Section 4.5.4.

Figure 3.1 traces the complete accounting: what is measured, what is estimated, what is generated by algebra, where each quantity is consumed, and how the loop closes.

## 3.6 Modelling Assumptions and Limitations

The relations of this chapter fall into two classes, and the distinction matters for how the results should be read.

**Relations that hold exactly.** For a payload on the thrust axis, the vertical displacement (3.11) of the centre of mass is collinear with the thrust, so the thrust produces no moment about the composite centre of mass; this is a consequence of geometry, not an omitted term. The increment (3.13) is exact for a rigid pair of point masses, and the yaw increment of (3.14) is exactly zero for a point mass on the yaw axis.

The remaining relations are approximations whose validity depends on the operating condition.

**The payload is assumed centred.** The model sets the horizontal attachment offset to zero. A residual offset $r_{xy}$ left by imperfect alignment produces a constant trim moment $m_P g\, r_{xy}$ and a roll/pitch inertia error $\mu r_{xy}^2$. At the $0.03\,\mathrm{m}$ attachment tolerance of the gripper and $0.30\,\mathrm{kg}$ these are $0.09\,\mathrm{N\,m}$, or $18\,\%$ of the roll authority, and $0.4\,\%$ of the increment. The first is supplied by the integral action of the autopilot's rate loop; the second is negligible.

**The arm is nominal, not measured.** Both optimisers use $r_z = -0.27\,\mathrm{m}$ for every flight. The realised arm is set by the height at which the vehicle grasps the

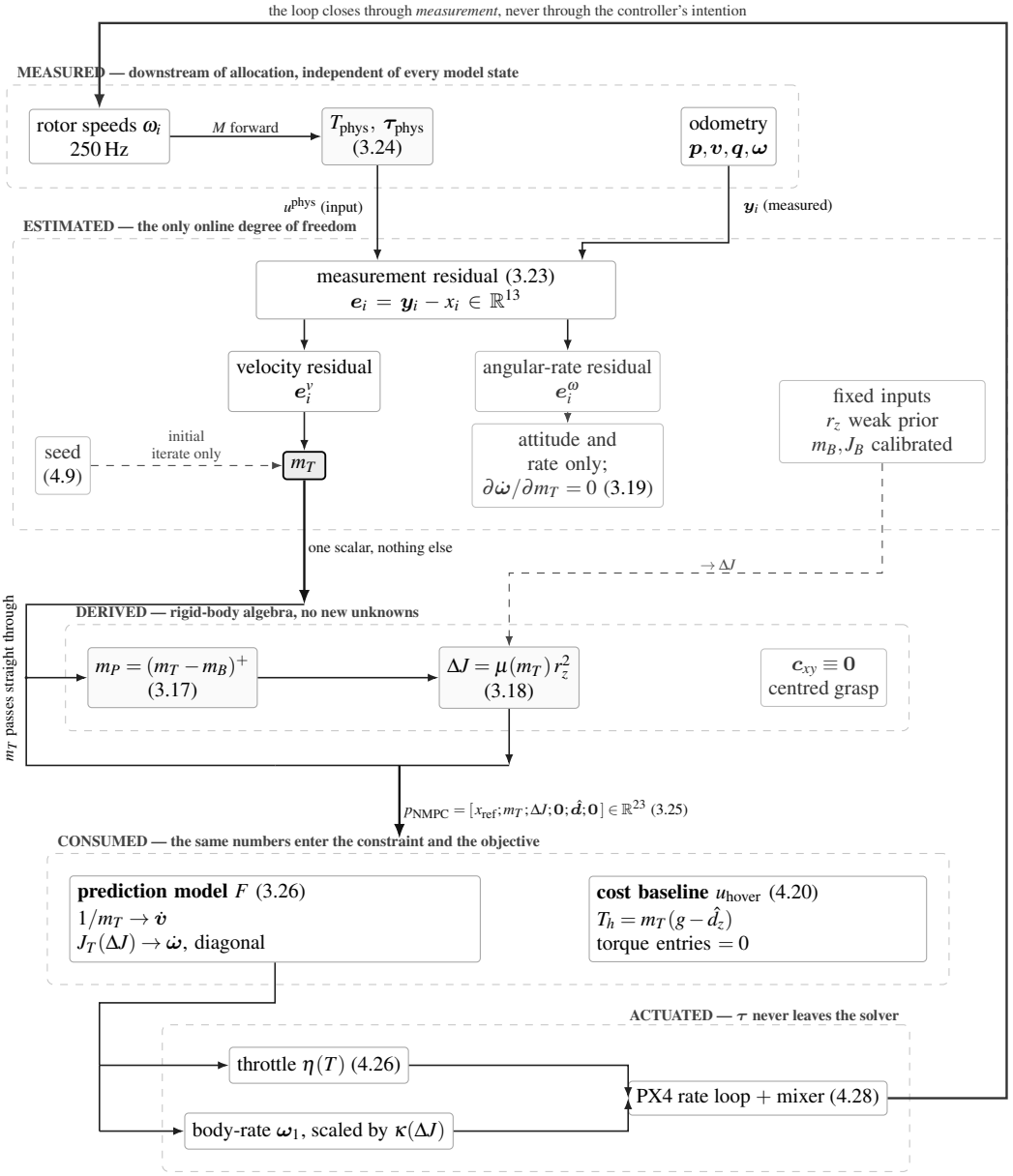



**Figure 3.1: How the payload parameter reaches the controller. The window's measurements and its known input meet in one residual** (3.23)**, whose velocity block is what identifies the single boxed decision variable; the angular-rate block constrains the vehicle's own attitude and rate but, by** (3.19)**, not the mass. Everything in the DERIVED band follows from $m_T$ by the algebra of Section 3.3 and introduces neither a new unknown nor a new sensor; the inertia increment vanishes by itself when the estimate returns to the bare airframe, so no release event is needed to remove it. The same increment, lagged by one solve** (3.20)**, is also written into the estimator's own rotational model as a known parameter. The horizontal-offset parameters are held at zero by the operating condition, not estimated. The loop closes through measured rotor speeds, never through the moment the solver intended. The residual specific force $\hat{d}$ in the controller's parameter vector is estimated from the same measurements but is not a payload parameter (Section 4.5.3).**

parcel, which is commanded in the autopilot's local frame whose vertical zero drifts from flight to flight; it varies between about 0.25 m and 0.30 m. Over that range the true increment for 0.30 kg lies between 0.0164 and 0.0236 kg m$^2$ against the modelled 0.0191 kg m$^2$ — an error of $-14$ to $+23$ percent, and a true inertia ratio between 2.15 and 2.66 against the modelled 2.34. Every result in this dissertation carries an arm error of this size, unmeasured on a per-flight basis.

**The parcel's own inertia is neglected.** The payload is treated as a point mass. A box of side 0.15 m has a central inertia of about $0.004\,m_P$, about 2 % of the roll/pitch increment and the only yaw contribution of a centred load.

**The airframe rotational model is approximate.** The calibrated airframe inertia $J_{xx} = 0.0142\,\mathrm{kg\,m^2}$ is about 60 % of the value in the simulator's model description, and a least-squares fit of bare-airframe pitch dynamics recovered an effective inertia 1.07–1.18 times the latter; rotor drag acting through the vertical offset of the rotor plane produces a velocity-dependent body moment that (3.9) does not contain; and a constant torque bias of order 0.01 N m appears in the reconstructed torque of an unloaded vehicle. These errors are the reason the mass is kept off the rotational channel (Section 3.4.2). They remain in the controller's prediction model, where they appear as an unmodelled moment that the rate loop absorbs.

**The estimator's inertia lags its mass by one solve.** The lagged treatment (3.20) evaluates the increment at the previous solution, so during a grasp or release the estimator's inertia trails the true value by one 0.1 s solve, and under sustained manoeuvre a biased mass estimate is fed back into the next window's rotational model. The treatment reduces the loaded mass bias but does not remove it (Section 5.7.2).

**The inertia is treated as constant across a transfer.** Equation (3.9) omits the $\dot{J}_T\boldsymbol{\omega}$ term that a time-varying inertia would contribute. The omission is exact on either side of a grasp or release and wrong only for the duration of the transfer itself; within the estimator the affected samples fall inside a single window and are absorbed as process noise.

**The rotor model is static.** The thrust coefficient $k_f$ is a constant, and is known to lose validity under large induced inflow during fast translation [34]; the mass channel recovers $m_T$ from an absolute thrust and inherits any error in $k_f$ in full. The yaw reaction of (3.24) is the steady-state term only, omitting the rotor angular-acceleration contribution $\sum_i J_r\dot{\omega}_i$, and rotor gyroscopic moments are not modelled. These terms are negligible in hover and mild manoeuvre but not during rapid yaw. Because $m_T$ is recovered from an absolute thrust, a relative error in $k_f$ passes one-for-one into $\hat{m}$; this sensitivity is analytical and has not been measured in the evaluated configuration (Section 5.9).

## 3.7 Chapter Summary

This chapter addressed the first research objective and the structural part of the second. For a rigid payload grasped on the thrust axis, the roll and pitch inertia increment is the reduced mass times the square of the vertical arm, so the composite mass is the only payload quantity that has to be identified online; the inertia follows from it algebraically and vanishes with it. The mass is identified on the translational channel, from a wrench reconstructed from measured rotor speeds, while the estimator's rotational model carries the inertia as a parameter lagged by one solve, so that the optimiser cannot use the mass to absorb rotational model error. The structure rests on stated assumptions — a centred grasp, a nominal arm and a calibrated rotor model — whose limits are collected in Section 3.6. What remains open is the numerical form of the

estimator and the controller, and how a release is recognised; Chapter 4 takes these up.